

# **DAY 11 - KALI LINUX SETUP**

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## **1. What is Kali Linux?**

Kali Linux is a Debian-based Linux distribution designed for cybersecurity work. It includes many tools used for authorized security testing, digital forensics, network analysis, and security research.

## **2. Why is Kali Linux used in cybersecurity?**

Kali provides a large collection of security tools in one environment. It is useful for learning, defensive analysis, authorized penetration testing, forensics, and security assessment in controlled labs.

## **3. What is a virtual machine?**

A virtual machine (VM) is a software-based computer that runs inside a physical computer. It has virtual hardware such as CPU, memory, storage, and networking and can run its own operating system.

## **4. What is a hypervisor?**

A hypervisor is software or firmware that creates and manages virtual machines. It allocates physical resources to VMs and keeps their virtual environments separated from one another.

## **5. What is VirtualBox?**

VirtualBox is a virtualization platform that allows users to create and run virtual machines on a host computer. It can be used to run operating systems such as Kali Linux in a controlled environment.

## **6. Explain Host OS and Guest OS.**

The Host OS is the operating system installed directly on the physical computer. The Guest OS is the operating system running inside a virtual machine managed by the hypervisor.

### **7. Why is using Kali inside a VM useful for beginners?**

A VM provides an isolated learning environment. Beginners can practice commands and security tools without directly changing the host operating system, and the VM can be restored using snapshots when something goes wrong.

### **8. What is a VM snapshot?**

A VM snapshot is a saved point-in-time state of a virtual machine. It can preserve the VM's disk state and configuration so the VM can be returned to that earlier state when supported by the virtualization platform.

### **9. Why should students create snapshots?**

Snapshots provide a recovery point before experiments or configuration changes. If a lab becomes misconfigured, the student can return to a known working state instead of rebuilding the VM from the beginning.

### **10. What does whoami display?**

whoami displays the username of the currently effective user in the terminal session.

### **11. What does hostname display?**

hostname displays the system's current host name, which identifies the computer on the network or within the local system configuration.

### **12. What does uname -a display?**

uname -a displays detailed system and kernel information, such as the kernel name, host architecture, kernel release/version, and related system information.

### **13. What does /etc/os-release contain?**

The /etc/os-release file contains identification information about the installed operating system, such as its name, ID, version, and related release details.

### **14. What does ip addr show?**

ip addr shows network interfaces and their assigned IP addresses, along with interface states and address information.

### **15. What does ip route show?**

ip route show displays the system's routing table, including routes to networks and the default route/gateway when configured.

### **16. What is sudo?**

sudo allows an authorized user to run a command with elevated privileges, usually as the root user. It is commonly used for administrative tasks while keeping normal work in a non-root session.

### **17. Why should elevated privileges be used carefully?**

Commands run with elevated privileges can change system-wide settings, files, permissions, or software. A mistake with sudo can therefore have a much larger impact than the same mistake made as a normal user.

### **18. What does sudo apt update do?**

sudo apt update refreshes the local package manager's information about available packages and repository metadata. It does not by itself upgrade installed packages.

### **19. Why should Kali Linux tools only be used on authorized systems?**

Security tools can discover, test, or interact with systems in ways that may affect availability, privacy, or security. Using them only on

systems you own or have explicit permission to test keeps the activity within an authorized and controlled scope.

**20. Explain the difference between ethical security testing and unauthorized access.**

Ethical security testing is performed with permission, a defined scope, and an agreed purpose such as finding and reporting security weaknesses. Unauthorized access or testing is performed without permission and can violate policies or laws and cause harm.